

The Bend Theorem

The power-sum mechanism of the composite profile law — the teeth are where the channels open

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0. STATEMENT

Let S_m be the Fermat surface of degree m , and let G be the SSvL line Gram. For a prime $p \mid m$, define the **bend constant**

$$\mathbf{b}(p) = \text{rank}_{\{F_p\}}(G) - 12(m-3).$$

Theorem (the Bend). $\mathbf{b}(p)$ depends only on p , through the divisibility $(p-1) \mid k$ of the pairing channels $k \in \{2, 4\}$:

- **$\mathbf{b}(p) = 0$ for every prime $p \geq 7$** — both channels closed, the profile law runs verbatim;
- **$\mathbf{b}(5) = 2$** — only the quadratic channel $k=4$ open: the two family-relative polarization radicals die, the global survives;
- **$\mathbf{b}(3) = 7 = 2 + 3 + 2$** — both channels open: two mass radicals, three moment-compatibility radicals, and two polarization radicals die; in every sector the family-relative copies die and the global combinations survive;
- ($p = 2$, even degree: both channels open and wild — the even half's separate kingdom, outside this document.)

Equivalently: composite-degree profiles obey the prime law per own prime with the corrections $\#m^2\text{-div} = 3m-16 + \mathbf{b}(p)$, $\#m\text{-div} = 3m^2-24m+59 - 2\mathbf{b}(p)$, and prime-power towers carry the bend as $2\mathbf{b}$ strays (field record below).

1. THE OBJECTS

The twelve shadow channels (3 families $\times$ 4 pencil directions $k, l, k-l, k+l$), each a function space on $\mathbb{Z}/m$. The form factors through the shadows at every $p \mid m$ (leak = 0, measured); the **shadow radical** has generic dimension $18 = 9$ (mass) + 6 (moment compatibilities) + 3 (polarizations), all CYCLIC across the family triangle (the hyperbolic SSvL pairing transports each identity to the neighbouring families' channels), with INTEGRAL coefficient frame (entries in $\{\pm 1, \pm 2\}$, certified). The entire radical is polynomial of degree ≤ 2 in t — beyond-degree-2 is zero in every measured kingdom — so the bookkeeping world is finite and complete. **$\mathbf{b}(p) = 18 - \dim(\text{radical mod } p)$.**

2. THE MECHANISM — THE PAIRING IS MADE OF POWER SUMS

Kernel structure (measured 78/78 blocks at $m=15$, integer level): every channel-pair kernel is a generalized circulant, $K(t, t') = \kappa(\alpha t + \beta t' \bmod m)$. Consequently the pairing of polynomial patterns $t^a \times t^b$ expands as an integer combination of power sums $\sum_t t^j$ with $j \leq a+b \leq 4$. Reduced mod p ($p \mid m$, where the polynomial calculus is canonical — see §7), the classical evaluation applies:

$$\sum_{t \in F_p} t^j \equiv -1 \text{ if } (p-1) \mid j, \text{ and } \equiv 0 \text{ otherwise.}$$

Channels: linear sector pairs through $k=2$; polarization sector through $k=4$. Open channels by prime: $p=3 \rightarrow \{2, 4\}$ both (measured row over $\mathbb{Z}/15$: $[0,0,1,0,1]$); $p=5 \rightarrow \{4\}$ only ($[0,0,0,0,2]$); **$p \geq 7 \rightarrow \text{none}$ ($p-1 \geq 6 > 4$): no polynomial detector can exist, hence $\mathbf{b}(p) = 0$ — the flat law is mechanism-complete, forever.**

3. THE THREE SECTOR CHAINS AT $p = 3$

(i) **Mass -2 [PROVEN, v95]**. The complete-Radon mass identity of F_3^2 (the four pencil slopes exhaust $P^1(F_3)$, and $4 \equiv 1$): one all-directions condition per family, 3 candidates - 1 overlap (sum of generic trinity = sum of Radon conditions; union rank 5) = 2 deaths, the dead named: the two cross-family $k \pm 1$ constant singletons.

(ii) **Linear -3 [conditions explicit; k=2 channel]**. Death conditions in the integral compat frame: $L_1 - L_4 + L_5 + L_6 = 0$, $L_2 - L_4 - L_5 - L_6 = 0$, $L_3 + L_5 + L_6 = 0$; survivors are the cross-family symmetric combinations (cleanest: $L_5 - L_6$, the $F_1 \leftrightarrow F_2$ symmetric moment pair). Detector functionals extracted from the annihilator with value-triples $(0,2,0)/(0,1,0)$; the channel is Σt^2 , open only at $p=3$.

(iii) **Polarization -2 [closed by identity coincidence + k=4 channel]**. Mod 3 the polarization pattern $(-2,-2,1,1) \equiv (1,1,1,1)$: the polarization identity and the all-directions quadratic identity are the same identity in char 3 (difference = $3(k^2 + l^2) \equiv 0$, literal). Survivor by formula: the global all-family combination; deaths: the two relatives.

4. THE FIVE-TOOTH — THE SAME ANIMAL, ONE CHANNEL

The polarization death conditions at $p=5$ are **identical** to $p=3$'s ($P_1 + P_3 = 0$, $P_2 - P_3 = 0$ in the integral frame; identical global survivor $(1,-1,-1) =$ the all-family pattern $(2,2,-1,-1)$). Every detector catching a polarization at $p=5$ is PURE quadratic, and all detector value-triples annihilate the survivor exactly. $b(5) = 2$ is the polarization bite through the only open channel, Σt^4 — not a separate mechanism. The founding "m=5 exception" of the odd-prime profile law was $b(5)$ biting at its own prime ($h_0 = 26 = 24 + 2$).

5. THE FLAT LAW, $p \geq 7$ — MECHANISM COMPLETE

All power sums $\Sigma(t \bmod p)^j$, $1 \leq j \leq 4$, vanish identically for $p \geq 7$. No pairing channel, no detector, no death: the radical reduces full, $b(p) = 0$. Field: 7, 11, 13 measured flat in every cell (own towers and as partners) — SEVEN $p \geq 7$ sightings itemized from files, zero exceptions.

6. THE FIELD RECORD

Profile-level seals $7/7 \cdot$ blind exponents $26/26 \cdot$ the bend formula $b(p) = h_0 - 12(m-3)$ exact on FIFTEEN field profiles (squarefree both sides, towers, gates) plus the two founding prime gates $\cdot$ prime-power towers: base-m floors verbatim ($m = 9, 25, 27$ with $2b$ strays; $m = 49$ with zero, the sealed falsifier that survived — $4/4$) $\cdot$ the necessary condition **bending requires $p \mid m$** PROVEN (no shadow factoring over Z).

7. HONEST LEDGER

PROVEN: the $p \mid m$ necessity; the mass-2 chain; the polarization identity coincidence (symbolic); the flat-law mechanism modulo the kernel expansion's general-m write-up. DEMONSTRATED STRUCTURALLY (measured, certified frames, detectors in hand): the kernel circulant structure ($78/78$ at $m=15$ — the general-m proof is the SSvL incidence rules, finite write-up); the linear-3 conditions; the channel table. MEASURED [FIRM]: the field record. The canonicity note: the polynomial calculus on Z/m exists mod p exactly when $p \mid m$ (moment weights wrap with step carries otherwise) — the teeth are where the structure exists, the second face of the $p \mid m$ necessity.

8. GRAVEYARD, WITH PRIDE

The one-tooth phrase (killed by its own table: $b(5)=2, 5 \nmid 6$) · the shadow-leak hypothesis (leak = 0 measured) · the integral-radical framing ($\sigma(\ker_Z G)$ rank 128 — its autopsy yielded the $p|m$ necessity) · the per-family symbol frame (the true frame is cyclic) · the foreign-prime witnesses at $m=15$ (non-canonical chop — its autopsy yielded the canonicity note) · "2+2+3" (overlapping filters; corrected 2+3+2) · one Auditor display-slip and one Constructor transpose, both self-caught. Each grave bought road.

9. PROVENANCE

Instruments: GRAM_MANOMETER_SURFACE_v4.py (ratified), gandalfsonrie.cpp (double-gated, checkpointed), session extraction scripts (radical formulas, symbol frames, detector functionals, kernel census). All numbers from files; seals before measurement; cross-review before any repo upload per house protocol. PMC.

References

1. M. Schütt, T. Shioda, R. van Luijk, *Lines on Fermat surfaces*. J. Number Theory **130**:9 (2010), 1939–1963. arXiv:0812.2377. (The line intersection rules behind the Gram.)
2. T. Shioda, *Some observations on Jacobi sums*. Advanced Studies in Pure Mathematics **12** (1987), 119–135. (The discriminant questions this campaign's surface theorems answer; the composite and even degrees are the territory this document's mechanism opens.)
3. E. Aljovin, H. Movasati, R. Villaflor, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628. (The published cell-by-cell elementary divisors; the campaign's gates.)

Companion documents in this repository: THE_WATERMARK_THEOREM (the prime-degree discriminant, proven) · THE_DOUBLE_LADDER_THEOREM (the surface structure) · THE_SWEET_LIE_THEOREM (the fourfold rank) · CHUCHIPACHI_FINDINGS_MASTER (the campaign record, graveyards included).